

CAPSTONE PROJECT

Enterprise Quality Engineering with Playwright & Generative AI

Target: ParaBank | parabank.parasoft.com | Playwright v1.4x+ | JavaScript (Node.js)



Project Overview

This capstone project engineers a production-grade, enterprise-level test automation ecosystem targeting **ParaBank** — a fully functional online banking demo application by Parasoft. Built using **Playwright v1.4x+** with **JavaScript (Node.js)**, the framework follows a strict **Page Object Model (POM)** architecture within a dual-project monorepo structure. Eight core banking services are automated with 15 test cases each, delivering a minimum of **120 automated test cases** executed across three browsers — Chromium, Firefox, and WebKit. The solution integrates a **GitHub Actions CI/CD pipeline**, Playwright HTML reporting, and **Generative AI paradigms** including Model Context Protocol (MCP) integration — demonstrating end-to-end quality engineering competency aligned with enterprise SDET standards.

Technology Stack

Component	Details
Framework	Playwright v1.4x+
Language	JavaScript (Node.js 20)
Browsers	Chromium, Firefox, WebKit
Architecture	Page Object Model (POM)
CI/CD	GitHub Actions
Reporting	Playwright HTML Reporter
AI Integration	Model Context Protocol (MCP)
Target Site	parabank.parasoft.com

Services Under Test

Eight banking services are selected to achieve comprehensive coverage of ParaBank's core functionality. Each service is implemented as a dedicated Page Object class with 15 test cases covering positive flows, negative scenarios, and edge cases.

1

Authentication

Covers user registration, login, logout, invalid credential handling, and session management. Tests verify that only valid users can access

2

Account Overview

Validates that account balances, account numbers, and account types (Checking/Savings) are displayed accurately.

3

Fund Transfer

Tests the core banking operation of transferring funds between accounts. Covers successful transfers, insufficient balance scenarios, invalid

4

Transaction History

Verifies that all completed transactions appear correctly in the account activity log. Covers date-range filtering, transaction type filtering,

5

Bill Payment

Covers the end-to-end bill payment workflow including payee selection, amount entry, account selection, and payment confirmation.

6

Loan Request

Tests the loan application feature where users request loans based on account standing. Covers approved and denied loan scenarios,

7

User Profile

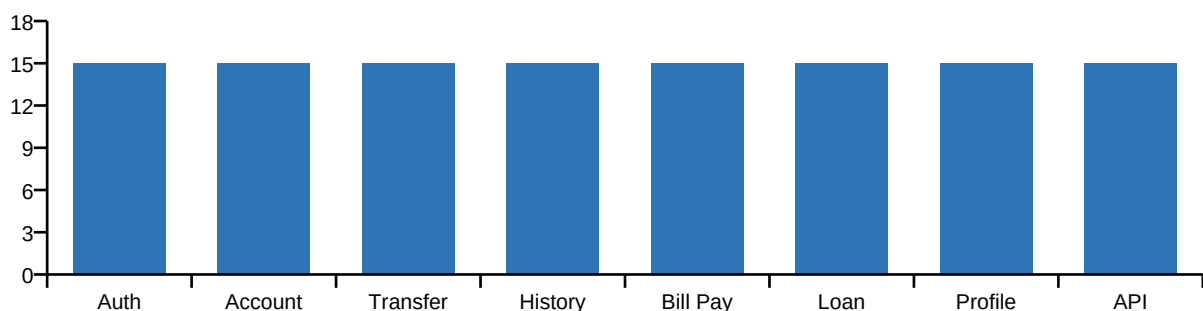
Validates account holder information update functionality including name, address, phone number, and contact details. Tests verify form

8

Internal API

Tests ParaBank's REST API layer directly using Playwright's request context. Covers login, account retrieval, transaction listing, fund

Test Case Distribution



Each of the 8 services has exactly 15 test cases — totalling 120 tests minimum.

Day-wise Delivery Plan

Day	Focus Area	Deliverable
1	Project setup & architecture	Config, BasePage, testData, CI scaffold
2	Authentication service	AuthPage.js + auth.setup.js (session saver)
3	Auth test cases	auth.spec.js — 15 tests complete
4	Account Overview service	AccountPage.js + 15 test cases
5	Fund Transfer service	TransferPage.js + 15 test cases
6	Transaction History service	HistoryPage.js + 15 test cases
7	Bill Payment service	BillPayPage.js + 15 test cases
8	Loan Request service	LoanPage.js + 15 test cases
9	User Profile service	ProfilePage.js + 15 test cases
10	Internal API service	ApiService.js + 15 API test cases
11	Project 2 — SVG Graphics Engine	DOM manipulation + shadow root tests
12	CI/CD finalization & full test run	GitHub Actions pipeline validated
13	Cleanup, README, GenAI/MCP documentation	Final submission ready

Evaluation Criteria

Criteria	Requirement	Status
Services	Minimum 8	8 defined
Test Cases	Minimum 120	120 planned
Architecture	Page Object Model	Enforced
Browsers	Chromium + Firefox + WebKit	All 3 configured
Daily GitHub Commits	Day 1 to Day 13	13 days planned
CI/CD Pipeline	GitHub Actions	playwright.yml ready
HTML Report	Playwright reporter	Configured
GenAI / MCP	Documented integration	Day 13
Dual Project	POM + SVG Engine	Both scaffolded

GitHub & Submission

Item	Detail
Repository	capstoneProject
Branch	main

Commit Format	Day{N}: <description> e.g. Day3: auth.spec.js — 15 tests complete
Day File Naming	Day1.md, Day2.md ... Day13.md
Reference Repo	github.com/aryan1403/capstoneProject